

DinkeyWeb (PHP)

Overview

DinkeyWeb (PHP) consists of a few core files.

- The defaults file, **dinkeywebdefaults.php**, which defines the DinkeyWebParams class in which you must set your site-wide default values. These are the values that will apply to most of the pages you protect (e.g., your SDSN and the type of dongle to use).
- The include file, **dinkeywebinc.php**, is the file that you must include in each of your protected web pages. You must NOT modify this file. It contains the `DDProtCheck()` function which you must call once from each page you protect passing it an instance of the DinkeyWebParams class.
- The runtime library is the library that is downloaded onto the user's PC and which performs the actual protection check. You must protect all runtime libraries with DinkeyAdd using the API method of protection. See the section "Runtime Libraries" later in this document for more information.

The DinkeyWeb (PHP) sample code also contains:

- **index.php** – an example of how to protect a web page. `DDProtCheck()` must be called before any output has been generated. If you need to set specific settings on a per-page basis, this can be done by setting those values in the instance of DinkeyWebParams that you create before calling `DDProtCheck()`. If the protection check fails, `DDProtCheck()` will not return to your protected web page but will redirect to the error page you define (see below).
- **dinkeyweberr.php** – an example of a page that will display protection check errors to the user. Note that you still need to call `DDProtCheck()` on the error page.

Setting up the DinkeyWeb (PHP) sample code on your web server

1. Create a directory on your web server (for example, call it **dinkeyweb**). Create a directory inside **dinkeyweb** called **libs**.
2. Edit **dinkeywebdefaults.php**:
 - `M_SDSN` should be set to your own SDSN.
 - `M_DongleType` should be set to the appropriate dongle-type constant for your dongle.
 - `M_LibrariesURL` should be set to the URL of the directory on your web server which will contain `dpWeb.dll` (i.e., the **libs** directory you created previously). For example, if your website is `www.example.com` then you must set `M_LibrariesURL` as follows:
`$M_LibrariesURL = "http://www.example.com/dinkeyweb/libs"`
 - `M_ErrorPageURL` should be set to the URL of the page that will be displayed when there is an error checking the dongle. For example:
`$M_ErrorPageURL = "http://www.example.com/dinkeyweb/dinkeyweberr.php"`
3. Upload the DinkeyWeb files to the **dinkeyweb** directory on your web server. So, your **dinkeyweb** directory should contain the following files and directories:
 - `index.php`
 - `dinkeywebdefaults.php`
 - `dinkeywebinc.php`
 - `dinkeyweberr.php`
 - `libs/dpWeb.dll`
4. Attach a dongle to your computer.
5. Open your web browser and visit the protected **index.php** that you uploaded (e.g., type "`http://www.example.com/dinkeyweb/index.php`" in the address bar).

6. You may have to give confirmation to allow the Java applet to run. It will perform a protection check and will then reload the protected **index.php** page which displays various properties about the dongle.

Note – if your php pages have non-ascii characters (e.g. for non-English languages) then some text editors (e.g. notepad) will save the files with a byte-order marker (BOM). This can cause display problems on some servers so it is recommended that you save your php files without the BOM.

Using the DinkeyWeb Self-Hosting Package

If you have chosen to use the DinkeyWeb Self-Hosting Package, then you will need to host the DinkeyWeb encryption and decryption programs (dpweb_encrypt.exe and dpweb_decrypt.exe) and the Java applet (DinkeyWeb.jar) on your own web server.

The encryption and decryption programs should not be accessible over HTTP (for security reasons) and so they don't need to be stored within your website structure.

The Java applet does need to be accessible over HTTP and therefore must be stored within your website structure.

You must edit **dinkeywebdefaults.php** as follows:

- `M_DinkeyWebSelfHostingPackage` must be set to TRUE.
- `M_EncDecPath` must be set to the full path on your server where the encryption and decryption programs are located (e.g., a path of the form “C:\dir\subdir” on Windows or “/dir/subdir” on Linux and Mac).
- `M_AppletURL` must be set to the URL of the directory on your web server that the Java applet is stored in (e.g., `http://www.example.com/dinkeyweb/appletdir`). Note that it should NOT contain the name of the applet file itself.

Runtime Libraries

A runtime library is automatically downloaded onto the user's machine in order to perform the protection check against the dongle. The available DinkeyWeb runtime libraries are listed below. You should protect (using DinkeyAdd) and upload the ones required for the platforms you wish to support.

dpWeb.dll	- Windows (32-bit)
dpWeb64.dll	- Windows (64-bit)
libdpWeb.so	- Linux (32-bit)
libdpWeb64.so	- Linux (64-bit)
libdpWeb.jnilib	- Mac OSX (32-bit)
libdpWeb64.jnilib	- Mac OSX (64-bit)

Note – if you protect each module separately in DinkeyAdd then they will have separate protection parameters. However, you can put all these modules into a folder and use wildcards in the Input Pathname of DinkeyAdd (Programs tab) to protect all these modules to a common set of protection parameters. This would make more sense if you were limiting the number of executions – for example.

AJAX

If you use AJAX in your web site or web application then you may wish to perform dongle checks from within your web pages after the page has loaded and passed the initial dongle check. This is possible using the DinkeyWeb AJAX API. You can call individual dongle checks or enable periodic background checks.

Under the directory for the server-side programming language you use (we will assume PHP in this documentation) you will find an AJAX directory. Please take some time to familiarize yourself with the files in that directory:

- **dwajax.js** – the client-side Javascript that drives DinkeyWeb AJAX. You must include it in your web pages. It exports some functions which you can call from your web page in order to use DinkeyWeb AJAX:

```
dwAjaxStartBackground(timeout, cbNotify, elemID, serverUri)
```

This function starts periodic background dongle checks in your web page.

Parameter	Description
timeout	Required. An integer that specifies the time in seconds between each background dongle check.
cbNotify	Required. A function to call each time a background dongle check completes. The callback function must be of the form: <code>function f(errno) {...}</code> where <code>errno</code> is the result of the protection check, 0 indicating success.
elemID	Required. A string specifying the id of an element on the containing page to use as the parent element for the DinkeyWeb Java applet. Note: The element can have CSS property <code>visibility</code> set to <code>hidden</code> but <i>must not</i> have CSS property <code>display</code> set to <code>none</code> .
serverUri	Required. The URI on the server at which the stub page (<code>dwajax.php</code>) is accessible over HTTP.

```
dwAjaxStopBackground()
```

This function stops the background dongle checks.

Parameter	Description
This function takes no parameters.	

```
dwAjaxProtectionCheck(cbNotify, elemID, serverUri)
```

Call this function to perform a single dongle check from your web page. It cannot be called while background dongle checks are enabled.

Parameter	Description
cbNotify	Required. A function to call each time a background dongle check completes. The callback function must be of the form: <code>function f(errno) {...}</code> where <code>errno</code> is the result of the protection check, 0 indicating success.
elemID	Required. A string specifying the id of an element on the containing page to use as the parent element for the DinkeyWeb Java applet. Note: The element can have CSS property <code>visibility</code> set to <code>hidden</code> but <i>must not</i> have CSS property <code>display</code> set to <code>none</code> .
serverUri	Required. The URI on the server at which the stub page (<code>dwajax.php</code>) is accessible over HTTP.

- **dwajax.php** – the server-side stub utilized by **dwajax.js**. It includes **dinkeywebinc.php** in the same way that your protected pages do. You may modify the path **dinkeywebinc.php** is included from depending on your set up. You can rename **dwajax.php** if you wish.
- **index.php** – a simple example page demonstrating how to call DinkeyWeb AJAX. Please familiarize yourself with this file and it's comments etc.